



敏實科技大學
MINTH UNIVERSITY OF SCIENCE AND TECHNOLOGY
人工智慧專業大學

MITUST · Nguyên lý xe tự lái · Tuần 5

Trang thứ hai của datasheet

規格表的第二層

Ba dòng ít ai đọc, nhưng hỏng là hỏng ở đó

三行很少人看，但出事都出在那裡

工具 | Công cụ :

ch-neural.github.io/MITUST-Principles-of-Self-Driving-Cars/week5/spec/

第五週 第 2 節 2026/10/08 (四) 10:10–11:00 CH Tseng

Phần 1 / 3

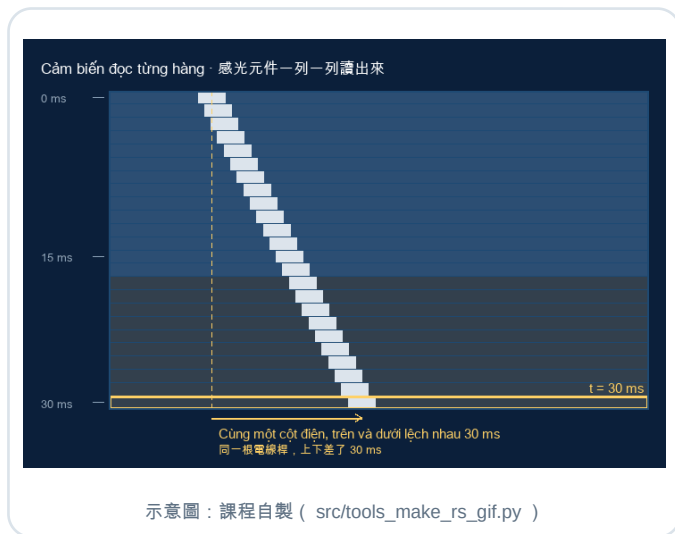
Camera: ảnh bị méo, bị đen hoặc trắng xoá, đèn đỏ thành tắt

相機：畫面會歪、會黑掉或白掉、紅燈會拍成沒亮

Kiểu màn trập · dải tương phản · nháy đèn LED

快門方式 · 動態範圍 · LED 閃爍

1 / 3 Rolling shutter



Màn trập cuộn: ảnh không phải chụp cùng một lúc, mà quét từ trên xuống

捲簾快門：一張照片不是同時拍的，是從上往下掃出來的

Cảm biến đọc từng hàng, hàng cuối chậm hơn hàng đầu vài chục mili giây. Trong khoảng đó xe và vật đều đã di chuyển, nên cột điện bị nghiêng, cánh quạt bị cong.

感光元件是一列一列讀出來的，最後一列比第一列晚了幾十毫秒。這段時間裡車子和目標都在動，所以電線桿會歪、螺旋槳會彎。

Xe tự lái sợ nhất: vật ở rìa ảnh bị lệch chỗ, mà thuật toán lại tưởng nó thật.

自駕車最怕的是：畫面邊緣的東西位置被扭掉，演算法卻以為那是真的。

2 / 3 Dynamic range



Dải tương phản động: cùng một khung vừa có chỗ tối vừa có chỗ chói

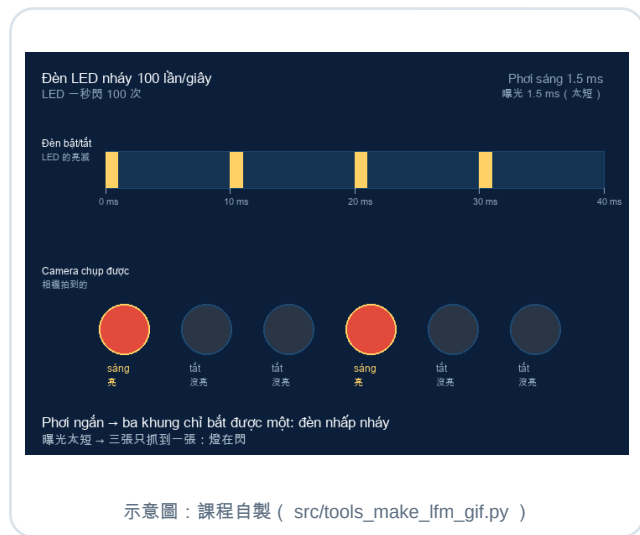
動態範圍：同一張畫面裡，又有很暗的地方又有很亮的地方

Ra khỏi đường hầm giữa trưa: trong hầm tối, ngoài hầm chói. Mắt người xử lý được khoảng 120 dB; AR0233AT ghi trên 140 dB, đó là lý do nó đắt.

中午開出隧道口：裡面很暗，外面很亮。人眼大約能處理 120 dB ；AR0233AT 寫的是 140 dB 以上，這就是它貴的原因。

Thiếu dải động thì hoặc là trong hầm đen kịt, hoặc là ngoài hầm trắng xóa — chọn một.
動態範圍不夠的話，不是隧道裡一片黑，就是隧道外一片白——只能選一個。

3 / 3 LED flicker



Đèn LED nhấp nháy: mắt người không thấy, camera thì thấy

LED 在閃：人眼看不出來，相機看得出來

Đèn tín hiệu và đèn hậu LED nhấp hơn một trăm lần mỗi giây, mỗi lần chỉ sáng 1/10 thời gian. Ban ngày phơi sáng rất ngắn nên camera hay bắt trượt.

紅綠燈和 LED 尾燈每秒閃一百次以上，而且一次只亮十分之一的時間。白天為了不過曝，快門很短，常常就剛好錯過。

Dòng LFM trên datasheet là để chữa đúng chuyện này.

規格表上那行 LFM，就是專門解決這件事的。

Phần 2 / 3

Lidar: laser loại nào, có hại mắt không, có xuyên mưa không

光達：雷射用哪一種、傷不傷眼睛、能不能看穿雨

Bước sóng · cấp an toàn laser · số lần phản hồi

波長 · 雷射安全等級 · 回波次數

905, 865 hay 1550 nm: chọn bước sóng là chọn đánh đổi

905、865 還是 1550 奈米：選波長就是選代價

Ba con số này đều nằm ở dòng «Laser Wavelength» trên datasheet.

這三個數字都寫在規格表的 Laser Wavelength 那一行。

Bước sóng 波長	Ai dùng 誰在用	Được lợi gì 好處	Trả giá gì 代價
905 nm 905 奈米	Phổ biến nhất 最常見	Cảm biến silic, rẻ 用矽做接收器，便宜	Tới võng mạc — công suất bị siết 直達視網膜，功率被壓住
865 nm 865 奈米	Ouster OS1 Ouster OS1	Cùng họ với 905 跟 905 同一類	Giống trên 同上
1550 nm 1550 奈米	Lidar tầm xa cao cấp 高階長距光達	Bị nước trong mắt hấp thụ → bắn mạnh hơn hàng chục lần → nhìn xa hơn 先被眼睛裡的水吸收 → 可以打強幾十倍 → 看得更遠	Cảm biến InGaAs, đắt hơn nhiều 要用銦鎵砷接收器，貴好幾倍

Một dòng số, ba chuyện: an toàn, nhìn xa, và giá.

一行數字，背後是三件事：安全、看多遠、多少錢。

An toàn mắt · 眼睛安全

Class 1 eye-safe: dòng này không có thì đừng lắp lên xe

Class 1 對眼睛安全：沒有這一行就不要裝上車

Nghĩa là dùng bình thường, nhìn thẳng vào nó cũng không hại mắt.

意思是在正常使用的情況下，就算直視它也不會傷到眼睛。

LASER

Laser Product Class	Class 1 eye-safe per IEC/EN 60825-1: 2014
Laser Wavelength	865 nm
Beam Diameter Exiting Sensor	9.5 mm
Beam Divergence	0.18° (FWHM)

Ouster OS1 datasheet, Rev7 / FW 3.2, p.2 · 節錄 · 讀取 2026/10/02

Đây không phải quảng cáo — đây là điều kiện để được bán.

這不是廣告詞，這是能不能賣的門檻。

Số lần phản hồi · 回波次數

«up to 2 returns»: một tia bắn ra có thể vọng về hai lần

「最多兩次回波」：一道雷射射出去，可能彈回來兩次

Tia xuyên mưa, lá cây, khói bụi rồi mới chạm vật thật. Chỉ nhận một lần vọng thì hạt mưa thành vật cản.
雷射穿過雨幕、樹葉、煙塵才打到真正的東西。只收第一次回波，就可能把雨滴當成障礙物。

# of Returns	up to 2
Return Order	Strongest to Weakest, Farthest to Nearest, and Nearest to Farthest

Ouster OS1 datasheet, Rev7 / FW 3.2, p.1 · 節錄 · 讀取 2026/10/02

Tuần 3 ta nói «tuyết bay cũng phản xạ» — dòng này chính là cách chữa.
第三週講過「飄著的雪也會反射」——這一行就是解法。

Mặt gương và vật phản quang: không về, về sai chỗ, hoặc quá chói

鏡面與反光材質：光回不來、回錯地方、或太亮

Bề mặt này làm gì 這個表面做了什麼	Thấy gì trên point cloud 點雲上看到什麼	Hiện xử lý thế nào 目前怎麼處理
Hất cả chùm đi, không về 整束反射走，一點都不回來	Thành lỗ: Kính toà nhà, cản mạ crôm, đường ướt 那一塊變成洞：玻璃帷幕、鍍鉻保險桿、濕路面	Radar bù: sóng mm gặp gương khác hắt 靠雷達補：毫米波碰鏡面的行為不一樣
Dội sang vật bên cạnh rồi mới về 先彈到旁邊的東西，再原路回來	Mọc ra xe không tồn tại ở phía sau gương (ghost point) 鏡子後面長出不存在的車（鬼點）	Nhiều lần phản hồi + lọc thời gian: ghost không ổn định 多次回波 + 時間比對：鬼點不會穩定移動
Phản xạ ngược (áo phản quang, biển báo) 回歸反射（反光背心、路牌）	Sáng quá mức, thành một quầng sáng 亮到爆表，變成一團光暈	Xem cường độ: phân bố khác hắt 看反射強度：分布和漫反射差很多

Không chiêu nào chặn được hết. Cách thật là ghép một tổ cảm biến yếu ở chỗ khác nhau.

沒有一招擋得住全部。真正的解法是湊一組「弱點不一樣」的感測器。

Thử nhanh · 小練習 (3 分鐘動手)

Mở công cụ, dùng công cụ nhỏ số ② để đổi «điểm mỗi giây» thành góc.

打開工具，用第②個小工具，把「每秒幾點」換算成角度。

1

Đặt 64 kênh, 1024 điểm mỗi vòng, 10 Hz — số điểm mỗi giây là bao nhiêu?

設成 64 線、每圈 1024 點、10 Hz——每秒點數是多少？

2

Phân giải ngang lúc đó là bao nhiêu độ? Ở 40 m hai điểm cách nhau mấy centimet?

這時候水平角解析度是幾度？40 公尺處兩個點相距幾公分？

3

Đổi sang 2048 điểm mỗi vòng: hai con số đó đổi thế nào?

改成每圈 2048 點：這兩個數字怎麼變？

Datasheet hay chỉ ghi «điểm mỗi giây» — con số oai nhất, nhưng không dùng thẳng được.

規格表常常只寫「每秒幾點」——那是最威風的數字，可是不能直接拿來用。

Phần 3 / 3

Hai cảm biến tốt, ghép lại vẫn sai

兩顆都是好的，合起來還是錯的

Hiệu chuẩn và đồng bộ thời gian

標定與時間同步

Hiệu chuẩn: nói cho xe biết mỗi con mắt ở đâu, nhìn hướng nào

標定：告訴車子每一隻眼睛裝在哪、朝哪看

1

Tham số trong

內參

Ống kính méo bao nhiêu

Tâm ảnh ở đâu

鏡頭扭曲多少

畫面中心在哪



2

Tham số ngoài

外參

Camera và lidar

cách nhau bao xa, lệch mấy độ

相機和光達

相距多遠、差幾度



3

Sai 1°

差 1 度

Ở 50 m lệch ngang gần 0,9 m

Đủ để nhầm làn

50 公尺處橫向差 0.9 公尺

足以搞錯車道

Rung, va chạm, thay kính chắn gió — đều làm lệch. Nên phải hiệu chuẩn lại định kỳ.

震動、碰撞、換擋風玻璃，都會讓它跑掉。所以要定期重新標定。

Đồng bộ · 時間同步

Đồng bộ thời gian: hai con mắt phải nhìn cùng một khoảnh khắc

時間同步：兩隻眼睛要看的是同一個瞬間

Lidar 10 Hz, camera 60 fps. Không có dấu thời gian chung thì lệch tới 50 ms — 40 km/h là hơn nửa mét.
光達 10 Hz、相機 60 fps。沒有共同時間戳，兩邊可能差 50 毫秒——時速 40 就超過半公尺。

Additional Programmability	• Multi-sensor phase lock • Queryable intrinsic calibration information: • Beam angles • IMU pose correction matrix	• Return ordering • Minimum range • Azimuth masking • Low-power standby mode
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Ouster OS1 datasheet, Rev7 / FW 3.2, p.3 · 節錄 · 讀取 2026/10/02

OS1 có dòng «multi-sensor phase lock» chính là để giải chuyện này.
OS1 規格表上那行 multi-sensor phase lock，就是在處理這件事。

Tiết 2 · 第二節小結

Mang ba câu này lên máy

帶三句話上機

Camera: màn trập cuốn làm méo vật đang chạy; thiếu dải động thì hầm đen hoặc trời trắng; không có LFM thì đèn đỏ có thể chụp thành tắt.

相機：捲簾快門會扭曲移動中的東西；動態範圍不夠，隧道裡黑或隧道外白；沒有 LFM，紅燈可能拍成沒亮。

Lidar: bước sóng quyết định được bắn mạnh tới đâu; Class 1 là ngưỡng bắt buộc; nhiều lần phản hồi là để sống sót trong mưa bụi.

光達：波長決定能打多強；Class 1 是門檻；多次回波是為了在雨和灰塵裡活下來。

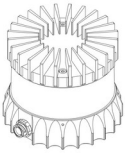
Hai cảm biến tốt chưa chắc ghép lại đúng: lệch 1° ở 50 m là gần 1 m; lệch 50 ms ở 40 km/h là hơn nửa mét.

兩顆好的感測器合起來不一定對：差 1 度，50 公尺就差近 1 公尺；差 50 毫秒，時速 40 就差半公尺以上。

Phụ lục · 附錄 (不計時)

Ouster OS1 — toàn bộ datasheet gốc (3 trang thông số)

Ouster OS1 — 廠商規格書原頁 (規格那三頁)



Ouster, Inc.
350 Third Ave.
San Francisco, CA 94110

OS1
Mid-Range High-Resolution Imaging Lidar
FIRMWARE VERSION: 3.2.x
HARDWARE VERSION: REV7.0

SUMMARY
The mid-range OS1 lidar sensor features 90 m range on a dark 10% target, a 42.4° vertical field of view, and high reliability for the most rugged conditions. The OS1 is designed for all-weather environments and use in industrial automation, autonomous vehicles, mapping, smart infrastructure, and robotics.

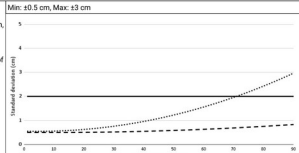
HIGHLIGHTS

- Configurable Minimum Range and Return Ordering
- Low Data Rate Profile now available with Dual Returns
- Camera-grade near-infrared and signal data
- Multi-sensor crosstalk suppression
- Ouster Studio for pointcloud evaluation
- Ouster SDK, ROS, and C++ drivers for SW development

OPTICAL PERFORMANCE

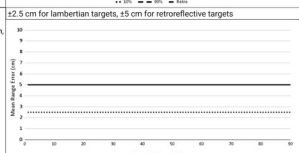
Range (R03 Lambertian reflectivity, 1024 @ 10 Hz mode)	170 m @ >90% detection probability, 100 kix sunlight
Range (10% Lambertian reflectivity, 1024 @ 10 Hz mode)	90 m @ >90% detection probability, 100 kix sunlight
Minimum Range	0.0 m (0.3 m optional, and 0.5 m default)
Vertical Resolution	32, 64, or 128 channels
Horizontal Resolution	512, 1024, or 2048 (configurable)
Rotation Rate	10 or 20 Hz (configurable)
Field of View	Vertical: 42.4° x 1.2° (x21.2° to 21.2°) Horizontal: 360°
Angular Sampling Accuracy	Vertical: ±0.01° / Horizontal: ±0.01°
False Positive Rate	1/10,000
Range Resolution	0.1 cm Note: For Low Data Rate Profile the Range Resolution = 0.8 cm
# of Returns	up to 2
Return Order	Strongest to Weakest, Farthest to Nearest, and Nearest to Farthest

Range Precision
(Typical on Lambertian and Retroreflective targets beyond 1 m, 1024 @ 10 Hz mode, 1 standard deviation)
Note: Precision is calculated based on the standard deviation of 100 measurements on a static target at a given range



Min: ±0.5 cm, Max: ±3 cm

Range Accuracy
(Typical on Lambertian and Retroreflective targets beyond 1 m, 1024 @ 10 Hz mode)
Note: Accuracy is calculated based on the error between the mean of 100 measurements on a static target at a given range and the true range



±2.5 cm for lambertian targets, ±5 cm for retroreflective targets

LASER

Laser Product Class	Class 1 eye-safe per IEC/EN 60825-1:2014
Laser Wavelength	905 nm
Beam Diameter Exiting Sensor	9.5 mm
Beam Divergence	0.18° (FWHM)

LIDAR OUTPUT

Connection	USB over gigabit Ethernet
Points Per Second	1,310,720 (32 channel) 2,621,440 (64 channel) 5,242,880 (128 channel)
Data Rate (megabits per second) (Low Data Rate Profile: 1 return, 1024 @ 10 Hz mode)	up to 11.83 Mbps (32 channel) up to 23.22 Mbps (64 channel) up to 43.29 Mbps (128 channel)
Data Rate (megabits per second) (Low Data Rate Profile: 2 returns, 1024 @ 10 Hz mode)	up to 23.22 Mbps (32 channel) up to 43.29 Mbps (64 channel) up to 85.24 Mbps (128 channel)

Data Rate (megabits per second) (Single Return Profile, 1024 @ 10 Hz mode)	up to 32.81 Mbps (32 channel) up to 64.26 Mbps (64 channel) up to 127.18 Mbps (128 channel)
Data Rate (megabits per second) (Dual Return Profile, 1024 @ 10 Hz mode)	up to 43.29 Mbps (32 channel) up to 85.24 Mbps (64 channel) up to 169.12 Mbps (128 channel)
Data Per Point	Range, Signal, Reflectivity, Near-infrared, Channel, Azimuth angle, and Timestamp
Timestamp Resolution	± 1 µs
Data Latency	± 10 ms
Data Integrity	End-to-End CRC that covers entire data packet

IMU OUTPUT

Connection	USB over 1000Base-T or 1000Base-T1
Samples Per Second	640
Data Per Sample	3 axis gms, 3 axis accelerometer
Timestamp Resolution	± 1 µs
Data Latency	± 10 ms
Additional Details	See Dense IAM 2068DHT datasheet for more details: https://invenSense.tdk.com/download/pdf/iam-2068DHT-datasheet/

CONTROL INTERFACE

Connection	HTTP API
Time Synchronization	Input sources: • IEEE1588 Precision Time Protocol (PTP), Accuracy: <1 ms error • NMEA SPP/RS485 UART message support • External PPS, Accuracy: <1 ms error • Internal 10 ppm drift clock, Accuracy: <20 ppm error Output sources: • Configurable 1 - 60 Hz output pulse
Lidar Operating Modes	• x 512 @ 10 Hz or 20 Hz • x 1024 @ 10 Hz or 20 Hz • x 2048 @ 10 Hz
Additional Programmability	• Multi-sensor phase lock • Queryable intrinsic calibration information • Beam angles • IMU pose correction matrix
	• Return ordering • Minimum range • Azimuth masking • Low-power standby mode

MECHANICAL/ELECTRICAL

Power Consumption	14 - 20 W • 16 W nominal • 20 W peak at startup if operating at 40 °C Note: Ouster recommends use of a power supply of no less than 30 W if using in cold conditions
Connector	Standard 1000BASE-T or Automotive Standard 1000BASE-T1
Operating Voltage	9.5 - 15 V • Suitable for 12 VDC to 24 VDC nominal systems • Not suitable for 48 V nominal battery based systems • Under-voltage WARNING level alert occurs at 9.0 VDC at the connector • Under-voltage ERROR level alert occurs at 9.0 VDC at the connector • Below 9.0 VDC at connector, sensor may shut down • Over-voltage conditions/alerts occur at 15 VDC at the connector • Over-voltage lockout onset at 15 VDC (±1 V) at the connector • Over-voltage lockout release at 15 VDC (±1 V) at the connector

Phụ lục · 附錄 (不計時)

Hesai AT128P — nguyên trang số tay (phần thông số)

Hesai AT128P — 使用手冊原頁 (規格那兩頁)

1.5. Specifications

1.5. Specifications

SENSOR

Scanning method	1-D rotating mirror (hybrid solid-state)
Number of channels	128
Instrumented range ①	0.5 to 260 m
Ranging capability	1 to 210 m (at 10% reflectivity)
Ranging accuracy	±5 cm (typical)
Ranging precision	3 cm (1σ)
Horizontal FOV	120°
Horizontal resolution ②	Far-field: 0.1° (10 Hz), 0.2° (20 Hz) Near-field: 0.2° (10 Hz), 0.4° (20 Hz)
Vertical FOV	25.4° (-12.5° to 12.9°)
Vertical resolution ②	Far-field: 0.2° Near-field: 0.8°
Frame rate	10/20 Hz
Return mode	Single Return: Last/Strongest/First Dual Return: Last and Strongest, Strongest and First

MECHANICAL/ELECTRICAL/OPERATIONAL

Wavelength	905 nm
Laser class	Class 1 Eye Safe
Ingress protection	IP6K7 & IP6K9K

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1.5. Specifications

Notes to specifications

① Instrumented range

- Actual measurement range, confined by the allocated Time of Flight (ToF) for each channel.
- For the point cloud data between 0.5 m and 1 m, ranging accuracy and precision cannot be guaranteed.

② Horizontal and vertical resolution

- Far-field measurement: >7.2 m
- Near-field measurement: 0.5 to 7.2 m

③ Dimensions and weight

- May be different for customized models.
- The mechanical drawings and data exclusively provided for customized models shall prevail.

④ Rated voltage range

Nominal voltages: 12 V and 24 V

⑤ Power consumption

- Typical value, not including accessories such as the connection box.
- The external power supply should be able to provide at least 45 W.

⑥ Valid point rate

- Defined as the point cloud data points (number of returns) generated per second.
- In **Single Return Mode**:
 - Given: horizontal FOV (120°), horizontal resolution (0.1° at 10 Hz), number of channels (128), and frame rate (10 Hz).
 - At 10 Hz, each channel generates 120/0.1 = 1200 points per frame; all channels generate 1200 × 128 = 153 600 points per frame, and 1 536 000 points (10 frames) per second.
- In **Dual Return Mode**, the point rate is twice that in **Single Return Mode**.
- The actual horizontal FOV of each mirror surface is slightly larger than 120°, so the actual point rate is slightly more than the valid point rate.

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Phụ lục · 附錄 (不計時)

onsemi AR0233AT — toàn bộ datasheet gốc (2 trang)

onsemi AR0233AT — 廠商規格書原頁 (全兩頁)

AR0233AT

Product Preview

1/2.5" 2.6-Megapixel HDR + LFM Automotive Sensor for ADAS & Viewing Systems

With new requirements being added to the New Car Assessment Program (NCAP), Advanced Driver Assistance Systems (ADAS) require sensors with higher functionality to meet the evolving standards for car safety. Automotive camera systems are performing more complex algorithms at higher vehicle speeds and in ever-changing lighting conditions. OEMs also want to offer new features for their customers by leveraging cameras for both viewing and sensing applications.

The AR0233AT image sensor delivers exceptional sensitivity for a wide variety of automotive applications. Built on a 3 μm Back Side Illuminated (BSI) pixel, the sensor provides greater than 140 dB of dynamic range and excels in low light conditions. It features an active array size of 2048 x 1280 with 1080p output at 60 frames per second. The new pixel technology includes LED Flicker Mitigation (LFM) while maintaining high dynamic range output, limiting the appearance of flicker from LED lighting and AC sources particularly important for camera monitor system (CMS) applications as well as machine vision applications.

The AR0233AT is available in multiple automotive-qualified package options including IBGA, CSP, and bare die. It is ASIL-B safety design compliant per ISO26262 and supports ASIL-B or higher safety-ratings for camera systems. The sensor is compatible with image signal processor (ISP) companion chips from ON Semiconductor as well as third party ISPs.

Features

- New 3 μm Dual Conversion Gain BSI Pixel Technology
- Multi-Exposure Mode for >140dB High Dynamic Range
- Full Resolution LED Flicker Mitigation with 120 dB High Dynamic Range
- >95 dB dynamic range from one exposure
- ASIL-B safety design, ISO26262 compliant
- Low-latency, low-power analog architecture
- Generative 3-Motion Compensation
- Adaptive Noise Reduction Filter
- Advanced Clocked Switching
- 4-Lane MIPI CSI-2 Interface
- 2048 x 1280 at 60 frames per Second
- 1920 x 60 at 60 frames per Second

All product information is provided under development. ON Semiconductor reserves the right to change or discontinue this product without notice.

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TBD
CASE TBD
TBO Number

ORDERING INFORMATION

Device	Package	Shipping
TBD	TBD	TBD

Specifications

- Optical format - 1/2.5"
- Active pixel – 2048 x 1280
- Pixel size – 3.0 μm
- Color filter array – RGB, RGCX & RCCB
- Interface – MIPI CSI-2, Parallel
- Max dynamic range – 140 dB @ exp., 120 dB (3-exp)
- Package options – IBGA, CSP, Bare Die

Applications

- Front View Camera (ADAS)
- In-Car Video Surveillance View and VVC
- ADAS – Viewing Fusion
- Camera Monitor Systems for Mirror Replacement

AR0233AT

PACKAGE DIMENSIONS

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ISSUE TBD

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